STANDARD V COVERINGS REFINE COVERINGS OF QUASI-COMPACT SEMI-SEPARATED SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3947, 3966] and proof(s) [[3968, 4038]]; this spec covers the complete environment instead (statement 3947-3966, proof span(s) [(3968, 4038)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original definition 0ETH: sources/stacks-chow/tags.txt, source lines 18517–18517. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0ETB: sources/stacks-chow/tags.txt, source lines 18511–18511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01J8: sources/stacks-chow/tags.txt, source lines 2005–2005. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $X \to Y$ be a quasi-compact morphism of schemes. The following are equivalent*

(1) *$\{X \to Y\}$ is a V covering,*
(2) *for any valuation ring V and morphism $g : \mathrm{Spec}(V) \to Y$ there exists an extension of valuation rings $V \subset W$ and a commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(W) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(V) & \longrightarrow & Y \end{array}$$

(3) *for any morphism $Z \to Y$ and specialization $z' \rightsquigarrow z$ of points in Z, there is a specialization $w' \rightsquigarrow w$ of points in $Z \times_Y X$ mapping to $z' \rightsquigarrow z$.*

Proof. Assume (1) and let $g : \mathrm{Spec}(V) \to Y$ be as in (2). Since V is a local ring there is an affine open $U \subset Y$ such that g factors through U. By Definition 0ETH we can find a standard V covering $\{U_j \to U\}$ refining $\{X \times_Y U \to U\}$. By Definition 0ETB we can find a j, an extension of valuation rings $V \subset W$ and a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(W) & \longrightarrow & U_j & \dashrightarrow & X \\ \downarrow & & \downarrow & \swarrow & \\ \mathrm{Spec}(V) & \longrightarrow & Y & & \end{array}$$

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We have the dotted arrow making the diagram commute by the refinement property of the covering and we see that (2) holds.

Assume (2) and let $Z \to Y$ and $z' \rightsquigarrow z$ be as in (3). By Schemes, Lemma 01J8 we can find a valuation ring V and a morphism $\mathrm{Spec}(V) \to Z$ such that the closed point of $\mathrm{Spec}(V)$ maps to z and the generic point of $\mathrm{Spec}(V)$ maps to z'. By (2) we can find an extension of valuation rings $V \subset W$ and a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(W) & & \longrightarrow & & X \\ \downarrow & & & & \downarrow \\ \mathrm{Spec}(V) & \longrightarrow & Z & \longrightarrow & Y \end{array}$$

The generic and closed points of $\mathrm{Spec}(W)$ map to points $w' \rightsquigarrow w$ in $Z \times_Y X$ via the induced morphism $\mathrm{Spec}(W) \to Z \times_Y X$. This shows that (3) holds.

Assume (3) holds and let $U \subset Y$ be an affine open. Choose a finite affine open covering $U \times_Y X = \bigcup_{j=1,\ldots,m} U_j$. This is possible as $X \to Y$ is quasi-compact. We claim that $\{U_j \to U\}$ is a standard V covering. The claim implies (1) is true and finishes the proof of the lemma. In order to prove the claim, let V be a valuation ring and let $g : \mathrm{Spec}(V) \to U$ be a morphism. By (3) we find a specialization $w' \rightsquigarrow w$ of points of

$$T = \mathrm{Spec}(V) \times_X Y = \mathrm{Spec}(V) \times_U (U \times_X Y)$$

such that w' maps to the generic point of $\mathrm{Spec}(V)$ and w maps to the closed point of $\mathrm{Spec}(V)$. By Schemes, Lemma 01J8 we can find a valuation ring W and a morphism $\mathrm{Spec}(W) \to T$ such that the generic point of $\mathrm{Spec}(W)$ maps to w' and the closed point of $\mathrm{Spec}(W)$ maps to w. The composition $\mathrm{Spec}(W) \to T \to \mathrm{Spec}(V)$ corresponds to an inclusion $V \subset W$ which presents W as an extension of the valuation ring V. Since $T = \bigcup \mathrm{Spec}(V) \times_U U_j$ is an open covering, we see that $\mathrm{Spec}(W) \to T$ factors through $\mathrm{Spec}(V) \times_U U_j$ for some j. Thus we obtain a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(W) & \longrightarrow & U_j \\ \downarrow & & \downarrow \\ \mathrm{Spec}(V) & \longrightarrow & U \end{array}$$

and the proof of the claim is complete. □